

USB-CAN Adapter/CANalyst-II Analyzer

Product Manual

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Chapter 1 Product Introduction

1.1 Overview

The USBCAN-2A/USBCAN-2C bus adapter is a CAN bus adapter with a USB 2.0 interface and a two-way CAN interface.

The CANalyst-II analyzer is a CAN analyzer with a USB2.0 interface and a 2-channel CAN interface. It has CAN bus protocol analysis functions and supports SAE J1939, DeviceNet, CANopen, iCAN, and custom high-level protocol analysis functions. It is also compatible with Chou Ligong's CANTest, CANPro, and ZCANpro software.

The USBCAN Bus Adapter/CANalyst-II Analyzer is a versatile tool for CAN bus product development, device testing and data analysis. It can act as a standard CAN node, allowing a PC to connect to a CAN network via the USB interface. This makes it useful in various fields such as bus test laboratories, industrial control, intelligent buildings and automotive electronics for data processing, acquisition and communication. The USBCAN/CANalyst-II analyzer is an ideal option for portable system users because of its compact size and easy installation. USBCAN-2A interface adapter device.

CAN bus is not isolated, directly powered by USB.

The USBCAN-2C interface adapter features an independent DC-DC power supply module for the CAN bus circuit and a high-speed signal isolation module for electrical isolation. This enhances the adapter's noise immunity, significantly improving system reliability in harsh environments. It is important to note that USB is isolated from CAN, but CAN1 is not isolated from CAN2.

The CANalyst-II analyzer provides complete isolation between USB, CAN1, and CAN2.

The USB_CAN TOOL software provided by the manufacturer allows direct configuration, sending and receiving of the CAN bus. Users can also write their own applications using the dynamic link library (DLL) and programming languages such as C++ Builder, C#, VC, VB, VB.NET, Delphi, LABVIEW, LabWindows/CVI,

Matlab, Python and QT routines. This facilitates the development of application software products for CAN systems.

When using the USBCAN adapter/CANalyst-II analyzer for secondary development, there is no need to understand the complex USB interface communication protocol.

1.2 Performance and Technical Indicators

- Protocol conversion between USB and CAN bus;
- USBCAN-2A, USBCAN-C, CANalyst-II analyzers are equipped with 2 channels of CAN interface;
- USB interface supports USB3.0, USB2.0, compatible with USB1.1 and OTG specifications;
- Support CAN2.0A and CAN2.0B protocols, support standard frames and extended frames;
- Support mutual transmission, CAN sending and CAN receiving;
- Support data frame and remote frame format;
- The baud rate of CAN controller is between 10Kbps-1Mbps, which can be configured by software;
- USBCAN-2A: CAN bus interface is not isolated;
- USBCAN-2C: The CAN bus interface uses high speed signal isolation and an isolated DC-DC power supply. The USB is isolated from the CAN, while CAN1 is not isolated from CAN2.
- CANalyst-II analyzer: the CAN bus interface adopts high-speed signal isolation and isolation of DC-DC power supply; USB, CAN1 and CAN2 are completely isolated from each other;
- Maximum flow: The system can receive and transmit more than 8500 frames per second per channel, and both channels can operate independently without interference.
- Internal CAN transmit buffer capacity: 20 frames/channel (automatic

retransmission when sending fails), CAN receive buffer capacity: 2000 frames/channel;

- USB bus direct power supply, no external power supply required; The voltage is 4.5V-5.5V, and the current is 100mA-200mA.

- USBCAN-2A series products, not isolated;

- USBCAN-2C series products, isolation module insulation voltage: 3000VDC;

- CANalyst-II analyzer, isolation module insulation voltage: 3000VDC;

- CAN1 channel built-in GDT components; CAN1 and CAN2 channels have built-in TVS transistors and common-mode inductors.

- Operating temperature: -20~85°C;

- Case size: USBCAN-2(A/C): 70*45*25mm; CANalyst-II analyzer: 104*70*25mm;

- Product compatibility: The library is compatible with Zhou Ligong's USBCAN-II/USBCAN-2E-U interface adapter.

1.3 Typical Applications

- Send and receive on the CAN bus network using the USB interface of a PC or notebook;

- Fast CAN network data collection and data analysis;

- CAN bus-USB gateway;

- USB interface to CAN network interface;

- Extend the network communication length of CAN bus;

- CAN network data monitoring in industrial sites.

1.4 Product Sales List

Serial Number	Name	Quality	Unit	Remarks
1	Main	1	pc	USBCAN Adapter/CANalyst-II Analyzer
2	Data package	1	pc	Including: manual, USB driver,

Serial Number	Name	Quality	Unit	Remarks
	(Download from official website)			Secondary development library file, secondary development sample program source code, USB_CAN TOOL test software and source code, baud rate detection tool, etc.
3	USB Cable	1	pc	present as a gift
4	3-inch screwdriver	1	pc	present as a gift

1.5 Technical Support and Service

We offer an unconditional return or exchange policy within 30 days of delivery. Additionally, we provide free maintenance for 5 years and lifetime maintenance and upgrade services.

For technical support and purchase information, please refer to

Email: zhcxgd@163.com

Technical support QQ: 3259558860

1.6 Product Selection

Model CAN	Number of CAN Channels	Isolation voltage Class (2500V)	Transparent Transmission Relay Function	High-level Protocol Analyzer Function
USBCAN-2A	2	Unisolated	Not Supported	Not Supported
USBCAN-2C	2	Industrial Isolated	Supported	Not Supported
CANalyst-II Analyzer	2	Industrial Isolated	Supported	Supported

Description:

1. All models support: USB-CAN Tool provided by the manufacturer (provide the original code), Chou-Li-Gong ZLG CANTest/CANpro/ZCANpro, JIYANG YANGDEN CANTools latest version of the software.
2. CANalyst-II analyzer: support the latest version of ZLG CANPro protocol analysis

platform software, support for high-level protocol analysis function.

3. USBCAN-2C/CANalyst-II analyzer: support CAN relay and transparent transmission function

Note:Please refer to the 'Relay, Transparent Transmission Function and Relay Software User's Manual (only supported by Supreme Edition).pdf' for instructions on the CAN relay function and transparent transmission function.

Chapter 2 Form Factor and Interface Description

2.1 Form Factor and Interfaces

The USBCAN/CANalyst-II Analyzer Interface Adapter has two external interfaces: a standard USB port and an 8-pin (6-pin for CANalyst-II analyzers) terminal block that provides the CAN bus interface.

The LED-PWR light, highlighted in red for CANalyst-II analyzers, indicates power.

The LED-CAN1 lamp, highlighted in yellow (blue for CANalyst-II analyzers), indicates the status of the CAN1 interface. The LED-CAN1 lamp blinks whenever CAN1 bus data is received or sent. (Upon plugging the USB-CAN bus adapter into the USB port, the system will perform a self-test and the LED-CAN1 lamp will blink once.)

The blue LED-CAN2 light (or red super bright for CANalyst-II analyzers) indicates the status of the CAN2 interface. The red LED-CAN2 light blinks whenever CAN2 bus data is received or sent. Upon plugging the USB-CAN bus adapter into the USB port, the system will perform a self-test and the LED-CAN2 light will blink once.)

The CANalyst-II analyzer features an additional SYS light that illuminates in blue when data transmission fails. If the data is not successfully sent within approximately one second, the transmission is cancelled and the light turns off.

Refer to the figure below for further details.



Figure 1 USBCAN series product outline diagram (specific to the specie)



Figure 2 CANalyst-II Analyzer Form Factor Diagram (in-kind)

2.2 Signal Definitions

USBCAN Series Products (USBCAN-2A/USBCAN-2C)		
	Name	Description
CAN1	R+	Terminating resistor R+. Shorting R+ to R- with a wire then the internal 120 ohm resistor will be tapped into the bus
	R-	Terminating resistor R-.
	CANH	CAN bus H signal.
	CANL	CAN bus L signal.
CAN2	R+	Terminating resistor R+. Shorting R+ and R- with a wire causes the internal 120 ohm resistor to be tapped into the bus
	R-	Terminating resistor R-.
	CANH	CAN bus H signal.

	CANL	CAN bus L signal.
Indicator light	PWR	Power indicator (always on)
	CAN1	CAN1 channel indicator (blinks when sending and receiving data)
	CAN2	CAN2 channel indicator (blinks when sending and receiving data)

CANalyst-II Analyzer Products (Supreme Edition Red, Silver, Linux)		
	Name	Description
CAN1	R1	Terminating resistor R1. dialed down to ON, the internal 120 ohm resistor is tapped into the bus
	R2	Terminating resistor R2. is connected in parallel with R1 and serves the same purpose. Two resistors are built-in for each channel.
	H	CAN bus H signal.
	S	Shielded line interface, if the communication line is shielded line can be connected to the shield, otherwise it can be grounded or not.
	L	CAN bus L signal.
CAN2	R1	Terminating resistor R1. dialed down to ON, the internal 120 ohm resistor is tapped into the bus
	R2	Terminating resistor R2. is connected in parallel with R1 and serves the same purpose. Two resistors are built-in for each channel.
	H	CAN bus H signal.
	G	The new version of the analyzer provides a signal ground interface for the CAN2 channel, which is internally connected directly to the signal reference ground.
	L	CAN bus L signal.
Indicator	PWR	Power indicator
	SYS	System status indication, normally off. Lights on when there is an error on the bus.
	CAN1	CAN1 channel indicator (blinks when sending and receiving data)
	CAN2	CAN2 channel indicator (blinks when sending and receiving data)

Working Mode:

CAN Send: The adapter receives data packets from the PC's USB interface, parses them, and saves them in the T-Buffer buffer. It then continuously reads the T-Buffer buffer to form a CAN message frame, which is sent to the CAN bus interface.

CAN Reception: The adapter receives data from the CAN network and stores it in the R-Buffer. When the host computer software requests to retrieve the data, the adapter sends it to the USB interface.

2.3 Factory Configuration

- 1) Software default CAN bus baud rate: 1Mbps; **Note: The baud rate of fault-tolerant CAN cannot exceed 125K.**
- 2) The software default acceptance mask register is 0xFFFFFFFF, which means no filtering, and CAN messages of any ID can be received.
- 3) The termination resistor can be set optionally: by shorting R+ and R- with a wire (or by toggling the corresponding switch for the CANalyst-II analyzer), the internal termination resistor of 120 ohms is used.

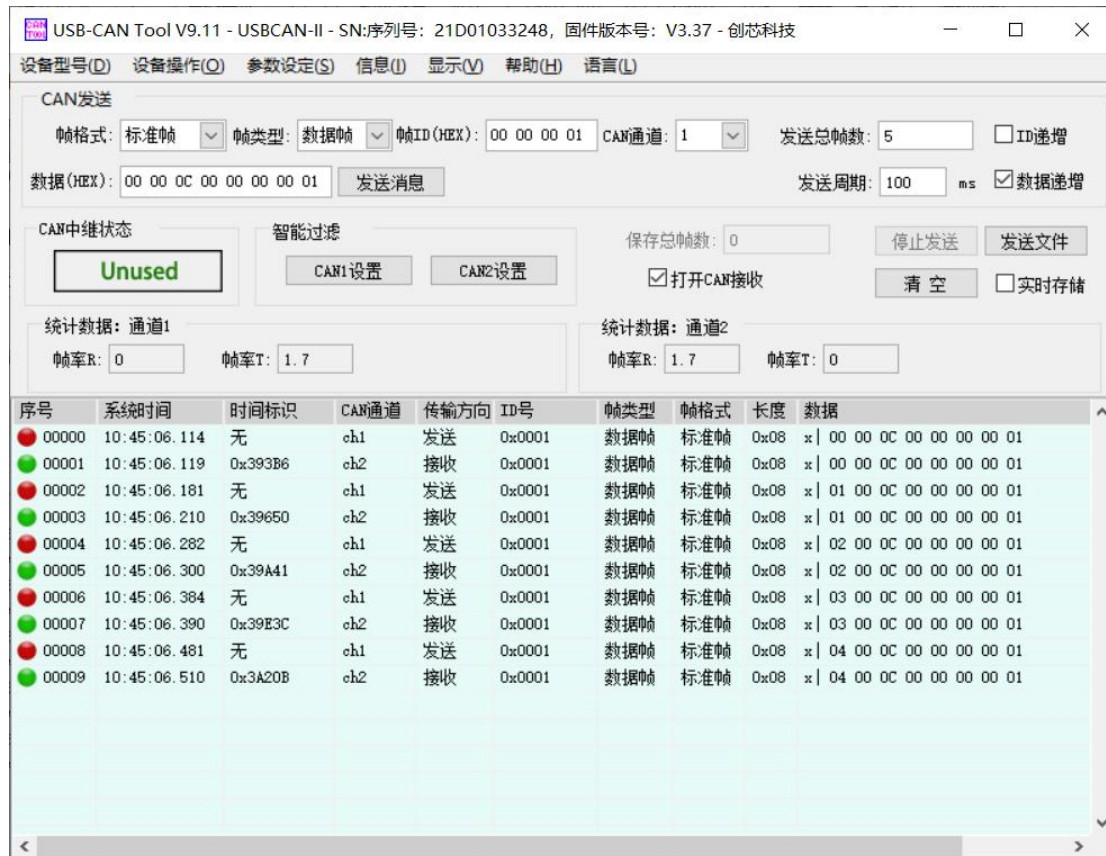
Note: Two 120-ohm termination resistors must be guaranteed on a normal bus or the CAN bus will not function properly.

Each CAN channel of the adapter is parameterized as above.

If terminating resistors are required, short R+ to R- with a wire for USBCAN, and turn the dip switch to the ON position for CANalyst-II analyzer.

Connect the device to the USB port of the PC via the USB connection cable;

Run the tool software USB-CAN Tool.exe test program as shown in Figure 4 below.\



3.3 Software Operation and Function Introduction

1、Select Model

USB_CAN-2(A/C) or CANalyst-II analyzer, menu "Device Model" -> USB-CAN2.0, check (system default).

2、Device Open/Close

Open the menu "Device Operation" -> "Start Device" to start the specified channel of the adapter, USB-CAN Tool is forced to start all channels of the device.

Open the menu "Device Operation" -> "Close Device" to close all channels of the running USB-CAN adapter.

3、CAN Parameter Configuration

Open the menu "Device Operation" -> "Start Device", the "Parameter Confirmation" dialog box will pop up. Configure the baud rate, filter setting and working mode in the dialog box. Generally, you only need to select the corresponding baud rate through the drop-down list, and the other parameters can be used by default. The CAN parameters of the two channels need to be configured separately.

Note: It is recommended to set the CAN parameters first before connecting the USB-CAN device to the bus. Please refer to "4. USB-CAN Tool Debugging Software Installation and User's Manual.pdf" for detailed instructions on parameter setting.

4、CAN Channel Selection

USB-CAN-2(A/C) has two CAN channels with index numbers 0 and 1 (i.e., CAN channel 1 and CAN channel 2).

When setting CAN parameters, please select the channel number.

5、CAN Baud Rate Setting

Modify it to the value required by the user. If communicating with external devices, the baud rate must be set to the same as that of the external CAN device.

6、Setting Message Filter

Users can set the filtering method and the values of message acceptance filter (ACR) and message mask filter (AMR) to filter the received CAN messages.

7、Operating Mode Settings

Normal operation mode, listen-only mode, self-test (loopback) mode.

Normal operation mode: The CAN module will appear on the CAN bus and can send and receive CAN messages.

Listen Only Mode: The module will appear on the CAN bus, but in a passive state. It will receive messages, but will not send messages or respond to messages. This mode can be used as a bus monitor as it does not affect data communication on the CAN bus.

Self-test (loopback) mode: This mode is used for the adapter to perform a self-test. It allows the CAN module to receive its own messages. In this mode, the CAN module transmit path is internally linked to the receive path. In this mode, a 'false' response is provided so that another node does not need to provide the response bit and the CAN message is not actually sent to the

CAN bus. CAN messages sent from the adapter will be received back by the adapter. (Our products are dual channel and can be tested directly for sending and receiving data between the two channels).

8、 Send CAN Message

When sending data you need to select Extended Frame/Standard Frame, Remote Frame/Data Frame, Frame ID, Data Length, Data and other information. In the CAN test software, enter the values in hexadecimal format in the ID and Data edit boxes, with a space between each value. When sending, take up to the first 4 values for ID and the first 8 values for data.

Note: In the time display of sent and received data, the time indicates the time when the PC displays the data and does not represent the actual time when the sending and receiving occurs, and there may be an error of up to 50ms between this time and the actual time. This is for reference only.

Note: The USB-CAN device has an automatic retransmission function to prevent data loss. If the device is not connected to the CAN bus or the baud rate does not match, it will not receive an answer signal. In this case, the device will automatically retransmit the data until it is received by other nodes on the CAN bus or until the USB-CAN device is powered down and restarted. Each channel has a data buffer capacity of approximately 20 frames. In case of unsuccessful transmission (with automatic retransmission), the host computer will call the transmit function and return 0 to indicate that the transmission has failed until the buffer data is sent normally. When using the timeout function, any unsuccessful frames within 1 second will be automatically cleared and not retransmitted.

9、 Send and Receive ID Format

Direct ID number format: the lowest bit (Bit0) of the ID is aligned with Bit0 of the ID byte. If ID=2, fill in 00 00 00 02 directly in the ID edit box. this format is intuitive and simple.

3.4 Self-Sending and Self-Receiving Test

Each CAN channel supports the self-test function.

The test steps are as follows:

(1) Connect the device to the USB port of the PC and run the USB-CAN Tool software;

(2) In the software menu "Device Operation", select "Start Device", then in the "Parameter Confirmation" dialog box, change the working mode to "Self-test (Loopback) Mode", and change the other parameters to "Self-test (Loopback) Mode".

(In the "Parameter Confirmation" dialog box, change the operating mode to "Self-test (Loopback) Mode", and use the default values for other parameters.

3) Click the "Send" button to send. Watch if you can receive back the CAN messages sent out, and these messages will be displayed in the data area.

The single-channel self-send and receive test cannot detect the CAN transceiver fault. For dual-channel CAN debuggers, it is recommended to perform the transmit and receive test directly between the two channels of CAN1 and CAN2, which can fully detect all faults of the CAN debugger.

USB-CAN-2(A/C), CANalyst-II, etc. have dual-channel CAN interfaces that can connect the CANH and CANL of the two channels accordingly to perform the transmit and receive test between the two channels. This test is more comprehensive and can react directly to the failure of the CAN transceiver, see the "5. Plug-in 1: USB-CAN Bus Adapter Test.pdf" instruction document.

Tip:

If the above operation does not work, please check that the steps are correct. If it still does not work, please contact and consult the manufacturer.

If no data is received after restarting the USB-CAN adapter, try restarting the USB-CAN Tool test software or restore the factory settings.

When using the adapter for debugging external CAN devices, please connect the CANH and CANL of the USBCAN to the corresponding CANH and CANL of the external CAN device, and leave the other wires unconnected.

3.5 Simultaneous Use of Multiple USB-CAN Devices

The USB-CAN Tool supports the connection of more than 10 USB-CAN bus adapters to the same computer at the same time, please refer to "12.Appendix 5: Multi-Card Operation Instruction Manual.pdf" for more details on this part.

Note: For more detailed instructions on how to use the USB-CAN Tool, please refer to: *"4. USB-CAN Tool Debugging Software Installation and User Manual.pdf"*.